ENGR 16100
Mars Cargo Rover (MACRO) Presentation of Competency (PoC)

The LC Harris Corporation is interested in evaluating the progress of potential design teams for the experimental Mars Cargo Mobility System (MACRO). In order to test competency, the Harris Corporation has identified a series of six tasks (four individual tasks and two integration tasks) relevant to the overall project. These tasks are to be performed at 'Presentation of Competency' (PoC) events. Tasks at PoC events target basic performance requirements identified in the Request for Proposal (RFP). Table 1 contains useful information for the performance of the individual tasks, and Table 2 has information relevant to the performance of the integration tasks. Table 3 contains general information pertinent to both the individual and the integration tasks. Teams should carefully consider what requirements or abilities will be required to succeed at these tasks.

Completion of individual tasks at a PoC event demonstrates the functionality of a subsystem. It is only necessary to successfully present each of these competencies once across all PoC events. Note that it may be desirable to present competencies more than once to demonstrate system integration. In the event that a previously demonstrated competency is failed at a later event, there is no effect on functionality scoring. All tasks are pass/fail – a MACRO is either competent or not. Each individual task is given a nominal value of 2 points.

To encourage the development of key functionality, LC Harris has designated combinations of related task integrations. Completion of an integration task demonstrates competency at both individual tasks, as well as added functionality for performing multiple tasks at once. Successful completion of an integration task carries a nominal value of 2 points **along with the possibility of earning points for an associated individual task.** For example, if a team successfully completes integration task 1/2 but has not yet earned points for completing individual task 2, that team would earn a total of 4 points (2 from the integration task, 2 from the individual tasks). Note that only individual tasks will be available at the first PoC event.

At each PoC event, teams will have a total of 9 minutes to complete all tasks that they wish to attempt. Teams will be allowed to spend a maximum of 2 minutes on any one task; the team will be allowed to make as many attempts as they wish on the task during the 2-minute time period. Teams will be allowed to interact with their robot between tasks; however, major alterations of the MACRO will not be permitted, and the team must complete all tasks within the 9-minute window. Instructional team members will be able to tell teams to move on to the next task or forfeit the remainder of the PoC event if they are deemed to be taking too much time.

LC Harris will evaluate this Presentation of Competency portion of the design proposal out of eight points in total; thus, successfully completing all individual tasks will meet expectations. Exceeding expectations will have a positive influence on the evaluation of the overall design proposal.

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Task		Other Task Notes
1	Demonstrate MACRO ability to traverse small obstacles while following a solid line.	MACRO must traverse a straight line track with various obstacles placed on it while carrying cargo.
2	Demonstrate MACRO ability to follow a dashed or broken line with a curve.	MACRO will be placed at the beginning of a dashed line on a track. The MACRO must follow the guiding line for a sufficient distance in a set amount of time.
3	Demonstrate MACRO ability to deposit cargo with the correct location and orientation.	MACRO is loaded with cargo at the start of a track outside of the drop-off zone. MACRO must move into the drop-off zone and deposit the cargo upright.
4	Demonstrate MACRO ability to scale different grades of terrain.	A block simulating both the upward and downward slopes of a hill of a certain grade will be placed along a straight line track. The MACRO must make it over the entire hill, then follow the track to its end.

Table 1 – General requirements of MACRO sub-systems

Integration Task		Other Task Notes
1/2	Demonstrate MACRO ability to navigate using guideline while traversing obstacles and following different line types.	MACRO is placed at start of guiding line; no cargo is needed. Must navigate following the line until end of the track is reached. Where options for line following are present, MACRO may choose path to follow.
3/4	Demonstrate MACRO ability to navigate over different grades of terrain and deposit cargo at correct location and orientation.	MACRO will be placed at beginning of track and loaded with one piece of cargo of team's choosing. Must successfully scale the upward and downward slope of a hill, then continue to drop-off zone and deposit cargo upright. Team will be told which drop-off zone to deposit in.

Table 2 – General requirements of MACRO integration tasks

Additional Notes
1. If the MACRO undergoes structural failure during any task (such as a part falling off) it fails the task automatically. The task may be restarted if time permits. 2. Once the MACRO has started an attempt on a task, teams may not interact with the MACRO in any way until the MACRO either completes the task or has clearly failed the task. Again, restarting is permitted if there is enough time. 3. All tasks will be performed on printed paper tracks. Departing the track or moving off of the paper is cause for automatic failure of the task. The task may be restarted if time permits. 4. Unless otherwise noted, stopping the MACRO is not important to the task.

Table 3 – Additional information related to PoC tasks

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